

YASIR JAFRI

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SUMMARY

Senior Backend Engineer with 5+ years of experience designing, building, and operating distributed systems and real-time data platforms. Specialises in event-driven microservices, real-time stream processing, and cloud-native architectures on AWS using Python and Go. Track record of owning systems end-to-end — from greenfield distributed system design to rearchitecting legacy platforms for scale, performance, and security compliance. Experienced in high-throughput Kafka-based pipelines, fault-tolerant service design, and PII-safe fintech infrastructure. Strong technical mentor with a bias for clean design, observability, and production reliability.

SKILLS

Languages: Python (primary), Go, Scala, Node.js (JavaScript), Java

Backend & Frameworks: FastAPI, Django, Flask, Gin, Express.js, Akka, Spark

Cloud & DevOps: AWS (Lambda, SNS, SQS, S3, ECS/EKS-style), Docker, Kubernetes, CI/CD, Terraform (basic)

Data & Messaging: PostgreSQL, MongoDB, Redis, ScyllaDB, Kafka, Elasticsearch, SQL Server, Clickhouse

Architecture & Practices: Event-driven microservices, real-time streaming, ETL pipelines, RESTful API design, system design, SOLID, TDD, Agile/Scrum, AI-assisted development, Generative AI, LLM, Agentic AI

EXPERIENCE

Senior Backend Engineer | **Appreciate** | Remote Jul 2024 – Present

- Architected and built a real-time stock price alert system from scratch as sole owner, tracking 15,000–20,000 stock symbols across a platform of 1M registered users (~10K DAU); delivered ~500 alerts/day with sub-2-second end-to-end notification latency using an event-driven, cloud-native stack on AWS (SQS, Lambda, SNS, Redis) with fault-tolerant fallback and batching to maintain reliability under cache unavailability.
- Rearchitected event broadcasting from direct API calls to an SNS-based fan-out model with dedicated SQS queues and Lambdas per platform (Appsflyer, CleverTap, MixPanel, Freshsales), decoupling services and improving delivery reliability across all downstream platforms.
- Implemented field-level AES encryption for all PII data — symmetric encryption at rest and asymmetric encryption in transit — in compliance with fintech regulatory requirements, covering user identity, financial, and sensitive profile fields.
- Identified and resolved latency bottlenecks across performance-critical APIs using distributed tracing; applied async-based concurrency to bring API response times from 2–3 seconds down to under 100ms.
- Hardened API security across critical endpoints following VAPT audits, implementing input validation and sanitisation, rate limiting, secure HTTP headers, XSS and CSRF protections, man-in-the-middle attack mitigations, and AES encryption for sensitive document uploads — closing all high-priority findings identified in the security review.
- Implemented Finite State Machine (FSM)-based parallel onboarding flows across Mutual Fund and US Stocks products; contributed PoC and core implementation alongside the feature owner.
- Built a UPI fund addition flow from scratch integrating Cashfree's payment gateway with Appreciate's Yes Bank account — implementing payment status callback polling, idempotent order management using internal order IDs, and a 3-retry failure recovery mechanism to ensure reliable fund transfers for end users.
- Follow Agile, TDD, and CI/CD practices; mentor engineers through design reviews, code reviews, and pairing sessions across a 20–30 person engineering team. Adopted AI-assisted development tooling including Cursor for coding and Axon AI (integrated in GitLab) for automated code reviews, improving team-wide review throughput and code quality.

Software Development Engineer I | **Gtropy (MapMyIndia)** | Gurugram Sep 2021 – Jul 2024

- Designed and operated a high-throughput Kafka-based GPS event pipeline for ~100,000 vehicles, consuming and compressing real-time telemetry data from 20GB down to 2GB per cycle by removing noisy signals, eliminating stoppage-period redundancy, marking trip boundaries, and storing data in a compact human-readable format — enabling downstream trip reconstruction, route optimisation, and fuel/toll cost analytics across states.
- Implemented a Trip Processing engine using Akka FSM and the Scala Actor system, enabling stateful, concurrent trip lifecycle management across 100,000+ vehicles with high throughput and fault isolation per trip.
- Optimised batch GPS data processing cron jobs using async and improved data structures for I/O-bound concurrency, reducing job execution time from 45 minutes to under 3 minutes.
- Led backend integration of a computer vision-based real-time driver monitoring system deployed across 10,000–15,000 vehicles — facial recognition on ignition, in-trip distraction and behaviour detection (phone use, smoking, head pose) — integrating Python backend services with ML inference models and edge devices, achieving 1–2 second end-to-end detection latency.

- Built a real-time social media scraping and NLP summarisation pipeline for Twitter, extracting actionable insights from transport-industry discourse and routing them into internal data stores.
- Migrated performance-critical legacy Django APIs to FastAPI, improving debuggability, throughput, and maintainability of high-traffic telematics endpoints.
- Designed and maintained RESTful APIs and backend services for fleet telematics and analytics features including driver behaviour, route analytics, and vehicle tracking.

Research Engineer & Technical Faculty | **iBotlab** | Remote Jan 2021 – Jun 2021

- Created technical content and prototypes in Data Analytics, Computer Vision, and Augmented Reality using Python.
- Led curriculum revamp that improved learner engagement and increased client acquisition by 30%, collaborating with stakeholders to align content with industry needs.

EDUCATION

M.Tech, Electrical Engineering | Jamia Millia Islamia, New Delhi GPA: 9.43 | Aug 2018 – Dec 2020

🏆 **Gold Medalist — Topper of the batch**

B.Tech, Electrical Engineering | Jamia Millia Islamia, New Delhi GPA: 8.61 | Aug 2013 – Jun 2017

Jun 2017 – Aug 2018: Preparation for M.Tech entrance examination (GATE).